Linear Algebra

Foundations Notes · Part 1

Scalars · Vectors · LI/LD · Span · Matrix Multiplication · Ax = b · Rank · Gaussian Elimination

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> Covers: Scalars · Vectors · Linear Dependence/Independence · Span (Filling R^n) · Matrix Multiplication (Coefficient/Column View) · System of Linear Equations · Rank · Gaussian Elimination

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1. Scalars

A scalar is a single number from a field (for GATE, almost always R — sometimes C or Q).

- Denoted by lowercase letters: `a, b, c, λ, k, α, β` .
- Operations: addition, multiplication, scalar inverse (if nonzero).
- Key role in LA: scalars are the "weights" used to scale vectors and build linear combinations.
- Field convention: for GATE the scalar field is always R (sometimes C in eigenvalue questions). The field determines what counts as a "linear combination", so it can in principle change LI/LD — but you won't see this nuance in GATE.

> Nuance: `0` is a scalar but also the zero of the field. Multiplying any vector by 0 gives the zero vector — this single fact is the gateway to half of linear dependence theory.

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2. Vectors

2.1 Definition

A vector in R^n is an ordered n-tuple of real numbers:

$$v = (v_1, v_2, \dots, v_n) \text{ or } \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

- Geometric view: an arrow from origin to the point (v_1, …, v_n) . Has magnitude + direction.
- Algebraic view: an element of the vector space R^n .
- Column convention: in LA we treat vectors as column vectors by default. Row vectors are transposes.

2.2 Core Operations

Operation	Definition	Geometric meaning
Addition	component-wise	Parallelogram / tip-to-tail
Scalar mult c·v	scale each component	Stretches when |c|>1, shrinks when |c|<1, flips when c<0
Zero vector 0	all components 0	Origin
Negative -v	(-1)·v	Same magnitude, opposite direction

2.3 Axioms a vector space must satisfy (10 axioms — memorize the gist)

Two closure axioms (closure under + and under scalar ·), plus 8 algebraic axioms:

1. Commutativity of +: u + v = v + u
2. Associativity of +: (u + v) + w = u + (v + w)
3. Additive identity: ∃ 0 with v + 0 = v
4. Additive inverse: ∃ −v with v + (−v) = 0
5. Associativity of scalar mult: a(bv) = (ab)v
6. Multiplicative identity: 1·v = v
7. Distributivity over vector sum: a(u + v) = au + av
8. Distributivity over scalar sum: (a + b)v = av + bv

GATE trap: If a set is missing the zero vector, it can't be a vector space / subspace. Always check 0 ∈ S first.

2.4 Linear Combination — the heartbeat of LA

Given vectors v_1, …, v_k and scalars c_1, …, c_k :

$$w = c_1v_1 + c_2v_2 + \cdots + c_kv_k$$

- Every concept ahead (span, dependence, rank, solution of Ax=b) is a question about linear combinations.

- Trivial combination: all $c_i = 0$ ⇒ result is 0 . This always works — it's the boring one.
- Non-trivial combination: at least one $c_i \neq 0$.

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3. Linearly Dependent & Independent Vectors

3.1 Formal Definition

Vectors $v_1, v_2, \dots, v_k$ are linearly independent (LI) if:

$$c_1v_1 + c_2v_2 + \cdots + c_kv_k = 0 \Rightarrow c_1 = c_2 = \cdots = c_k = 0$$

i.e., the only way to combine them to get zero is the trivial way.

They are linearly dependent (LD) if there exists at least one non-trivial combination giving 0 .

3.2 Equivalent Characterizations (any of these ⇔ LD)

1. Some v_i can be written as a linear combination of the others.
2. One vector is redundant — removing it doesn't shrink the span.
3. The matrix $[v_1 \mid v_2 \mid \dots \mid v_k]$ has rank $< k$.
4. If $k = n$ (same as dimension), $\det([v_1 \mid \dots \mid v_n]) = 0$.
5. The homogeneous system $Ax = 0$ has a non-trivial solution.

3.3 Quick-Fire Rules (memorize cold — these save minutes in GATE)

Situation	Status	Reason
Set contains the zero vector	LD	$1\cdot 0 + 0\cdot v_2 + \dots = 0$ is non-trivial
Single nonzero vector $\{v\}$	LI	$cv = 0 \Rightarrow c = 0$
Single zero vector $\{0\}$	LD	(above rule)
Two vectors, one a scalar multiple of the other	LD	$v_2 = kv_1$ ⇒ $kv_1 - v_2 = 0$
Two vectors, neither a multiple of the other	LI	—
More than n vectors in R^n	Always LD	Can't have > n LI vectors in n-dim space
Subset of LI set	LI	Independence is hereditary downward
Superset of LD set	LD	Dependence is hereditary upward
Orthogonal set of nonzero vectors	LI	Take dot product with v_i — only c_i survives

> Trick: Before doing any computation, count. If you see 3 vectors in R^2, 4 in R^3, 5 in R^4 — instantly LD. Don't compute.

3.4 How to Check LI/LD — methods ranked by speed

1. Inspection — multiples? zero vector? count > dim?
2. Determinant — only when matrix is square (k = n). $\det \neq 0 \Leftrightarrow$ LI .
3. Row reduce $[v_1 \mid v_2 \mid \dots \mid v_k]$ to echelon form. Count pivots: - pivots = $k \Rightarrow$ LI - pivots < $k \Rightarrow$ LD
4. Solve $Ax = 0$ — only trivial solution ⇒ LI.

3.5 Nuances often tested

- LI is a property of the set. Saying "a single vector v is LI" is just shorthand for "{v} is LI" — which boils down to $v \neq 0$. The interesting LI/LD questions start at 2+ vectors.
- A set of LI vectors may become LD when you add a vector (you introduced a redundancy).
- LI does not require orthogonality. Nonzero orthogonal set ⇒ LI, but LI ⇏ orthogonal.
- LI/LD is invariant under reordering of the vectors. Order doesn't matter.

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4. Span — "Filling" the Space R^n

4.1 Span Definition

$$\text{span}\{v_1, \dots, v_k\} = \{c_1v_1 + \cdots + c_kv_k : c_i \in \mathbb{R}\}$$

The set of all possible linear combinations of the given vectors.

4.2 What does span look like geometrically?

Vectors	Span (assuming nonzero & non-trivial)	In R^3 it's...
1 vector	a line through origin	a line
2 LI vectors	a plane through origin	a plane
3 LI vectors	all of R^3	the whole space
2 LD vectors	a line	a line (collapse!)

The "size" of the span is the number of LI vectors, not the number of vectors you started with.

4.3 When do vectors span R^n? (the filling condition)

Vectors $v_1, \dots, v_k$ span R^n iff the matrix $A = [v_1 \mid \dots \mid v_k]$ has rank n (full row rank).

- Need at least n vectors to span R^n . (Fewer ⇒ impossible.)
- Exactly n LI vectors span R^n and form a basis.
- More than n vectors can span R^n but then they're LD (some are redundant).

4.4 Basis & Dimension

- Basis = a set that is (1) LI and (2) spans the space. Both conditions required.
- Dimension of R^n = n . Every basis of R^n has exactly n vectors.
- Standard basis of R^n : $e_1 = (1,0,\dots,0), e_2 = (0,1,0,\dots,0), \dots, e_n = (0,\dots,0,1)$.

4.5 Tight relationships — keep these on the tip of your tongue

For k vectors in R^n :

- $k < n$: can be LI, but cannot span R^n.
- $k = n$: LI ⇔ spans R^n ⇔ basis ⇔ det ≠ 0 ⇔ invertible. (The big equivalence!)
- $k > n$: spans R^n possible, but always LD.

> GATE Tip: In any "is it a basis?" question with $k = n$, just compute the determinant. Nonzero ⇒ basis. Done.

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5. Matrix Multiplication — Coefficient (Column) View

5.1 The Four Views of AB

Let A be $m \times n$, B be $n \times p$. Then AB is $m \times p$.

View	What it says	When to use
Entry view (standard)	$(AB)_{ij} = \Sigma_k a_{ik}b_{kj}$ (row · column)	Computing single entries
Column view (coefficient view) ★	Column j of AB = A · (column j of B) = linear combination of columns of A with weights from column j of B	Solving $Ax = b$; understanding rank
Row view	Row i of AB = (row i of A) · B = linear combination of rows of B with weights from row i of A	Row space arguments
Outer product view	$AB = \Sigma_k (col_k A)(row_k B)$ — sum of rank-1 matrices	SVD, low-rank approximations

5.2 Column View in Depth — the one you must internalize

For $Ax = b$:

$$Ax = x_1 a_1 + x_2 a_2 + \cdots + x_n a_n$$

where a_j is the j-th column of A .

Ax is a linear combination of the columns of A, with the entries of x as the coefficients.

> This single reinterpretation is the most important sentence in early linear algebra. Burn it in.

Immediate consequences:

- $Ax = b$ has a solution ⇔ $b \in span\{columns\ of\ A\}$ = column space $C(A)$.
- $Ax = 0$ has a non-trivial solution ⇔ columns of A are LD.
- If columns of A span R^m , then $Ax = b$ is solvable for every b .

5.3 Properties — keep handy

- Associative: $(AB)C = A(BC)$ ✓

- Distributive: $A(B+C) = AB + AC$ ✓
- NOT commutative: $AB \neq BA$ in general. (They may not even have compatible sizes.)
- $(AB)^T = B^T A^T$ (order flips)
- $(AB)^{-1} = B^{-1}A^{-1}$ (order flips, when both invertible)
- $AI = IA = A$
- $A \cdot 0 = 0$, but $AB = 0$ does NOT imply $A = 0$ or $B = 0$. (Zero-divisor trap.)
- $AB = AC$ does NOT imply $B = C$ in general. You can left-cancel A only when A has full column rank (left-invertible) — for square A that means A is invertible.

5.4 Size sanity check (GATE one-liners)

$A_{m\times n} \cdot B_{n\times p} = C_{m\times p}$. Inner dimensions must match. Always check this first in any matrix problem — you'll catch typos and trick questions.

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6. System of Linear Equations: $Ax = b$

6.1 Two Pictures

Row picture: each equation = a hyperplane in R^n . Solution = intersection of all hyperplanes.

Column picture: find weights $x_1, \ldots, x_n$ so the columns of A combine to give b . (Column view from §5.2.)

> Switch between these views fluently. The column view is what GATE tests most.

6.2 Three Cases of Solvability

Case	Condition (in terms of rank)	Geometric meaning
No solution (inconsistent)	$\text{rank}(A) \neq \text{rank}([A\|b])$	Hyperplanes have no common intersection; $b \notin C(A)$
Unique solution	$\text{rank}(A) = \text{rank}([A\|b]) = n$ (# unknowns)	Single intersection point
Infinitely many solutions	$\text{rank}(A) = \text{rank}([A\|b]) < n$	Intersection is a line/plane/…

Master formula (consistent case only): $\#\text{ free variables} = n - \text{rank}(A)$.

- 0 free variables ⇒ unique solution.
- $k \geq 1$ free variables ⇒ ∞ solutions, parametrized by k parameters.

6.3 Homogeneous Systems: $Ax = 0$

- Always consistent — $x = 0$ is always a solution (the trivial solution).
- Has non-trivial solution ⇔ $\text{rank}(A) < n$ ⇔ columns of A are LD ⇔ (if square) $\det(A) = 0$.
- Solution set is a subspace of R^n called the null space $N(A)$, with $\dim N(A) = n - \text{rank}(A)$ (nullity).

6.4 Non-homogeneous: $Ax = b$

- General solution = (any particular solution) + (general solution of $Ax = 0$)

$$x_{gen} = x_p + x_h$$

- Not a subspace (doesn't pass through origin unless $b = 0$).

6.5 Square Systems (A is $n \times n$) — the cleanest case

The following are equivalent:

1. A is invertible.
2. $\det(A) \neq 0$.
3. $\text{rank}(A) = n$ (full rank).
4. Columns (and rows) of A are LI.
5. Columns of A span R^n.
6. $Ax = 0$ has only the trivial solution.
7. $Ax = b$ has a unique solution $x = A^{-1}b$ for every b.
8. 0 is not an eigenvalue.

> The "Invertible Matrix Theorem" — memorize this chain. GATE recycles it constantly.

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7. Rank

7.1 Definitions (all equivalent)

- Max number of linearly independent rows (row rank).
- Max number of linearly independent columns (column rank).
- Number of pivots in any echelon form of A.
- Dimension of the row space = dimension of the column space.
- Order of the largest non-vanishing minor (determinant of submatrix).

Theorem: row rank = column rank (always). ✓ (Why we just say "rank".)

7.2 Bounds & Basic Properties

For A of size $m \times n$:

- $0 \leq \text{rank}(A) \leq \min(m, n)$.
- $\text{rank}(A) = 0 \Leftrightarrow A = 0$ (zero matrix).
- $\text{rank}(A) = \text{rank}(A^T)$.
- Full row rank: $\text{rank}(A) = m$. Full column rank: $\text{rank}(A) = n$. Full rank (square): both at once.

7.3 Key Rank Inequalities (GATE-bait)

- $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$
- $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$ ★ — very commonly tested
- $\text{rank}(AB) \geq \text{rank}(A) + \text{rank}(B) - n$ (Sylvester's inequality; n = inner dim)
- If A is $m \times n$ of rank r, then $\text{nullity}(A) = n - r$. (Rank–Nullity Theorem)
- $\text{rank}(A^T A) = \text{rank}(A A^T) = \text{rank}(A)$ (over R)
- Multiplying by an invertible matrix doesn't change rank: $\text{rank}(PA) = \text{rank}(AQ) = \text{rank}(A)$ when P, Q invertible.

7.4 How to Compute Rank

1. Row-reduce A to echelon form.
2. Count non-zero rows = number of pivots = rank.

> Echelon form (not RREF) is enough — saves time. RREF only when you also need solutions.

7.5 Rank-Based Reading of $Ax = b$ (one-glance method)

Compare $r = \text{rank}(A)$ and $r' = \text{rank}([A|b])$:

Comparison	# solutions
$r \neq r'$	None (inconsistent)
$r = r' = n$	Unique
$r = r' < n$	Infinite (with $n - r$ free vars)

> Always do this first before any actual solving — it tells you what to aim for.

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8. Gaussian Elimination (to solve $Ax = b$)

8.1 Three Elementary Row Operations (ERO)

Symbol	Operation	Effect
$R_i \leftrightarrow R_j$	swap two rows	reorder equations
$R_i \to cR_i$ ($c \neq 0$)	scale a row	multiply equation by c
$R_i \to R_i + cR_j$	add multiple of another row	linear combo of equations

> EROs preserve the solution set. That's the whole point.

8.2 The Procedure

1. Form the augmented matrix $[A \mid b]$.
2. Forward elimination: use EROs to get row echelon form (REF): - All zero rows at the bottom. - Each leading (pivot) entry is to the right of the one above it.
3. Check consistency: - Any row of the form $[0\ 0\ \dots\ 0 \mid c]$ with $c \neq 0 \Rightarrow$ no solution. Stop.
4. Back substitution (or continue to reduced REF with leading 1's and zeros above each pivot, then read solution directly).

8.3 Echelon Form vs Reduced Echelon Form

- REF: zeros below each pivot. Useful for finding rank/consistency.
- RREF: zeros above and below each pivot, pivots are 1 . Useful for writing the explicit solution.

8.4 Pivots, Basic & Free Variables

- A column with a pivot ⇒ corresponding variable is a basic (pivot) variable.
- A column without a pivot ⇒ corresponding variable is a free variable (parameter).
- (For consistent systems) # free variables = n − rank(A).
- This same number `n − rank(A)` is the nullity of `A` (dimension of the null space `N(A)` = solution space of `Ax = 0` , which is always consistent).

8.5 Reading off solution from RREF (recipe)

1. If any row is `[0…0 | nonzero]` → no solution.
2. Else, assign each free variable a parameter (`t, s, …`).
3. Express each basic variable in terms of the free ones using its pivot row.
4. Write `x = x_p + t_1 v_1 + t_2 v_2 + …` (particular + homogeneous parts).

8.6 Worked-style template (commit this pattern)

```
Augmented:  [A | b]
   ↓  forward elimination (EROs)
REF:       [U | c]      ← check: any 0…0 | nonzero?  → No solution
   ↓  back-sub  (or RREF + read)
Solution:   x = x_p + Σ t_i · n_i
                └particular┘   └basis of N(A)┘
```

8.7 Common Pitfalls / Tricks

- Never scale a row by `0` . (Invalidates the operation.)
- Never do `R_i → R_i + 0·R_j` — pointless.
- Keep one operation per step; mixing two EROs in one step is where students miscount.
- When elements get ugly (fractions), it often signals a bad pivot order — try a row swap.
- Partial pivoting (always pick the largest-magnitude entry as pivot) is a numerical tip, useful in computational questions.
- After elimination, always sanity-check by plugging at least one component back into one original equation.

8.8 Reading off `rank` , `nullity` , # solutions in one pass

Once you've reached REF of `[A | b]` :

- `rank(A)` = pivots in the `A` portion.
- `rank([A|b])` = pivots in the whole augmented matrix.
- Compare per §7.5 — instantly classify the system.

8.9 Why elimination works (the conceptual anchor)

Every ERO is equivalent to left-multiplying by an invertible elementary matrix `E` . So forward elimination is the composition `E_k … E_2 E_1 A = U` . Each `E_i` is invertible, so `rank` and the solution set of `Ax = b` are preserved.

If `A` is square and elimination proceeds with non-zero pivots and no row swaps, you've effectively factorized `A = LU` (LU decomposition — preview for later).

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9. Quick GATE Cheat-Sheet (last-minute glance)

- More vectors than dimensions ⇒ LD. (Don't compute.)
- Square matrix: `det = 0` ⇔ LD columns ⇔ singular ⇔ `Ax = 0` has non-trivial sol ⇔ rank < n.
- `Ax = b` solvability: compare `rank(A)` and `rank([A|b])` .
- # free variables = `n − rank(A)` (for consistent systems).
- `rank(AB) ≤ min(rank A, rank B)` — favorite trap.
- Rank doesn't change under invertible multiplications, transpose, or EROs.
- Column view of `Ax` : linear combination of A's columns with x's entries as weights.
- Two parallel/proportional rows in `A` ⇒ rank drops by 1.
- Zero row/column ⇒ rank loses 1.
- Standard basis `{e_1, …, e_n}` is LI, spans `R^n` , and is orthonormal.
- Homogeneous system always consistent; non-trivial iff `rank < n` .
- Determinant only defined for square matrices; rank is defined for any size.

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10. Mental Models / "Why I should care"

1. Vectors = arrows you can stretch and add.
2. Linear combination = recipe for cooking new vectors out of old ones.
3. Span = pantry — everything you could cook with the ingredients you have.
4. LI = no wasted ingredients — each contributes a flavor you can't get from the others.
5. Basis = minimal complete cookbook for the space.
6. Rank = how many genuinely independent ingredients you actually have.
7. `Ax = b` : "Can I cook `b` using the recipes encoded by A's columns? If yes, how many ways?"
8. Gaussian elimination = simplifying the recipe until the answer is obvious — without changing what you're cooking.

End of Part 1. Up next: Determinants → Inverse → Eigenvalues/Eigenvectors → Diagonalization → LU/Cholesky → Special matrices.